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MECHANICAL ENGINEERING / AUTOMOTIVE RESTORATION

PROJECT PORTFOLIO

USF / TAMPA, FL



2014 SUBARU BRZ / 2013 DODGE DART / IEEE EXO

2014 SUBARU BRZ

RESTORE / REFINE / DRIVE











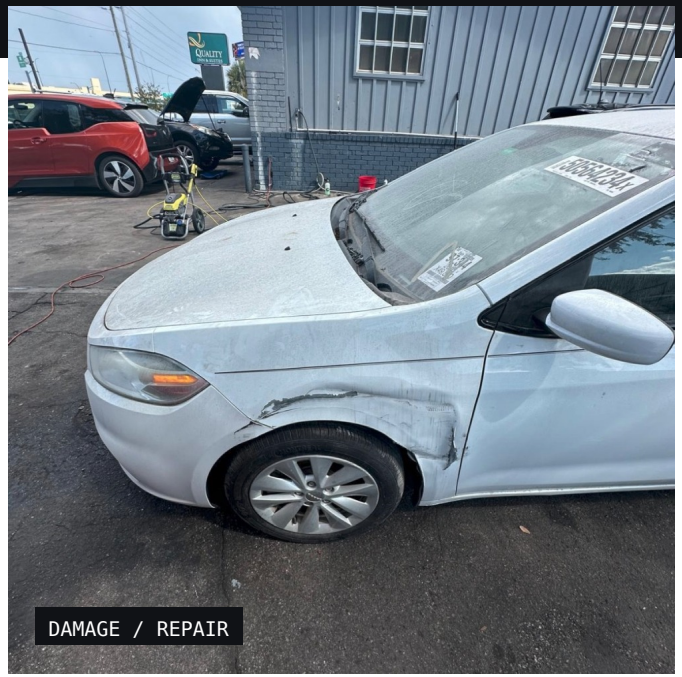
WHAT CHANGED

Body repair, replacement parts, paint preparation, reassembly, cleaning, and manual-transmission troubleshooting.

2013 DODGE DART

NON-RUNNING / DRIVABLE


AUCTION / BEFORE


DAMAGE / REPAIR


AFTER

FROM A $500 NON-RUNNER TO A DRIVABLE CAR

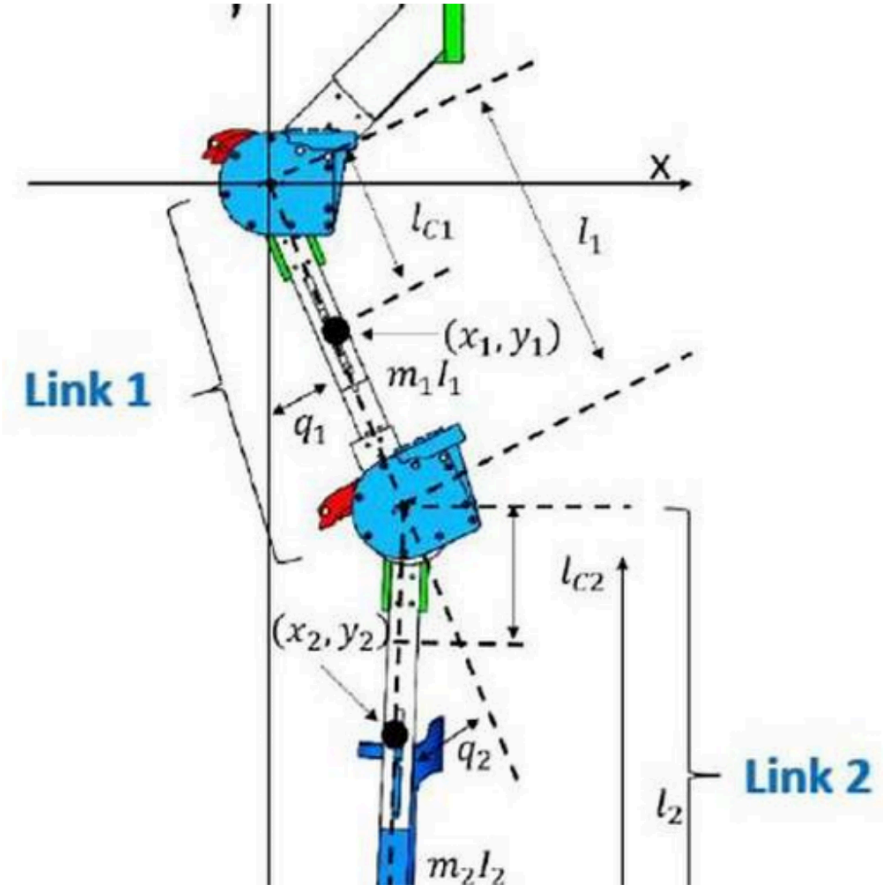
The team purchased the Dart through a dealership-auction connection. Work centered on diagnosing the transmission and shifting problem, repairing or replacing components, and restoring the exterior before resale.

PROJECT NOTES

PURCHASE	**$500**
ISSUE	**Transmission / shift linkage**
RESULT	**Returned to drivable condition**

MECHANICAL LINKAGE REFERENCE

2 LINK MODEL / HIP AND KNEE ROTATION / SAGITTAL PLANE



STUDY MODEL. LINK 1 REPRESENTS THE THIGH. LINK 2 REPRESENTS THE SHANK. Q1 AND Q2 ARE HIP AND KNEE ROTATION.

DESIGN USE

REFERENCE

The study reduces one leg to 2 rigid links with rotational hip and knee joints.

CHECK IN CAD

Hip center, knee center, link length, actuator clearance, cuff location, and required range of motion.

MECHANICAL OUTPUT

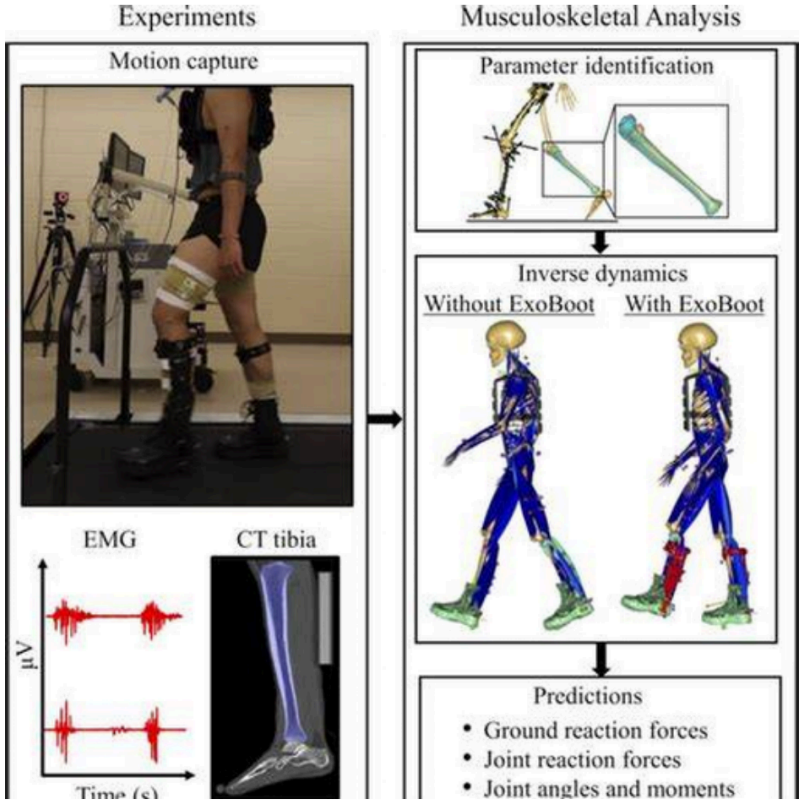
Joint locations and link geometry that can be carried into the unilateral CAD assembly.

LIMIT

This figure is reference geometry. Final dimensions come from project measurements, selected hardware, and the CAD assembly.

OPENSIM SIMULATION

MOTION DATA / MODEL SCALING / JOINT ANGLES / JOINT MOMENTS



PUBLISHED WORKFLOW SHOWING MOTION CAPTURE, SUBJECT DATA, MUSCULOSKELETAL MODELING, INVERSE DYNAMICS, AND JOINT LEVEL OUTPUTS.

PLANNED USE

INPUT

Marker trajectories, force data, and subject measurements from the gait trial.

MODEL

Scale the musculoskeletal model to the participant. Run inverse kinematics. Run inverse dynamics.

OUTPUT

Hip, knee, and ankle angles. Joint moments. Peak timing through the gait cycle.

MECHANICAL USE

Angles set motion and clearance checks. Moments support torque targets. Timing gives Controls the assistance window.

RANGE OF MOTION

Check that the joint assembly can reach the measured angles without binding or hard contact.

TORQUE TARGET

Use joint moment results as an input for actuator sizing and assistance targets.

TIMING

Locate peak demand in stance and swing. Controls can use the same timing window.

TRIAL COMPARISON

Compare walking speed, dual task, stairs, and added load using the same output set.

ANALYSIS TO CAD

LAB DATA / OPENSIM / REQUIREMENTS / MECHANICAL DESIGN

01 COLLECT MOTION AND FORCE DATA

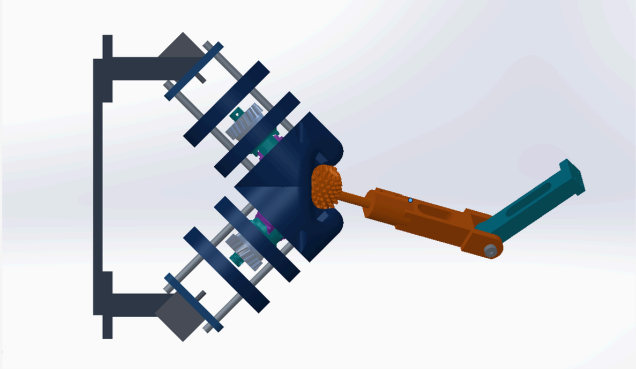
02 SCALE THE PARTICIPANT MODEL

03 RUN INVERSE KINEMATICS AND INVERSE DYNAMICS

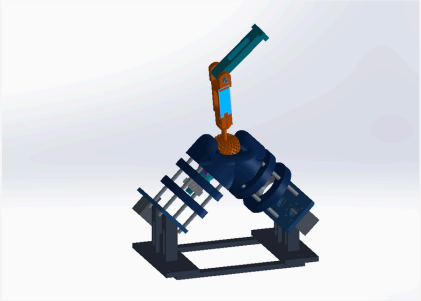
04 RECORD JOINT ANGLE, MOMENT, AND TIMING TARGETS

05 CHECK CAD GEOMETRY AND ACTUATOR REQUIREMENTS

CAD View 1



CAD View 2



CAD concept views of the exoskeleton linkage and spherical-gear assembly.

MECHANICAL CHECKS

JOINT ALIGNMENT

Keep the device hip and knee axes close to the user joint centers through the intended motion.

CLEARANCE

Check actuator, link, cuff, and frame clearance through the modeled range of motion.

FIT

Check cuff location, strap path, adjustment range, and slippage before powered testing.

VERIFICATION

OpenSim is a design input. Physical fit checks, structural analysis, bench testing, and hardware testing are still required.

MECHANICAL TEAM OUTPUT

Joint locations, link lengths, actuator mounts, cuff geometry, clearances, BOM items, fabrication drawings, and assembly notes.

OPENSIM OUTPUT

Joint angles, joint moments, and gait timing used to set motion limits, torque targets, and control timing.